

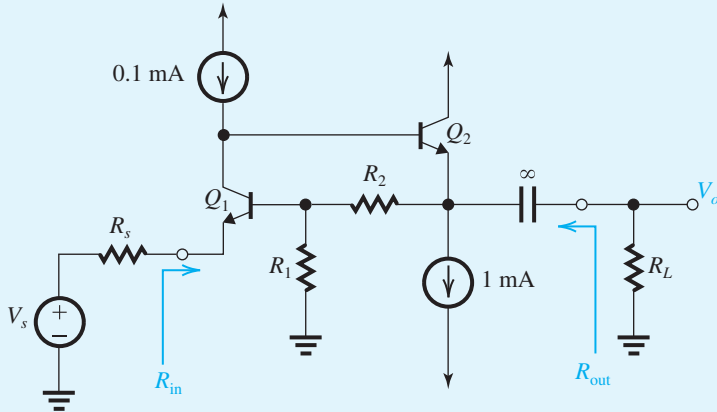


Figure P11.29

- (b) Use the expression for $A\beta$ derived in Example 11.3 to find the value of the loop gain for the case $\mu = 1000$, $R_{id} = 100 \text{ k}\Omega$, $r_o = 1 \text{ k}\Omega$, $R_s = 100 \text{ k}\Omega$, and $R_L = 10 \text{ k}\Omega$. Hence determine the value of the closed-loop gain A_f .
- (c) By what factor must μ be increased to ensure that A_f is within 1% of the ideal value of 10?

D 11.28 Consider the series–shunt feedback amplifier of Fig. 11.8(b) that is analyzed in Example 11.2.

- (a) If $R_1 = 1 \text{ k}\Omega$, what value should R_2 have to obtain a closed-loop gain whose ideal value is 5 V/V?
- (b) If $g_{m1} = g_{m2} = 4 \text{ mA/V}$, $R_{D1} = R_{D2} = 10 \text{ k}\Omega$, and the MOSFET's r_o is very large, use the expression for $A\beta$ derived in Example 11.2 to find the value of $A\beta$ and hence determine the closed-loop gain A_f .

***11.29** In the series–shunt feedback amplifier shown in Fig. P11.29, the devices operate with $V_{BE} = 0.7 \text{ V}$ and have $\beta_1 = \beta_2 = 100$. The input signal V_s has a zero dc component. Resistances $R_s = 100 \text{ }\Omega$, $R_1 = 1 \text{ k}\Omega$, $R_2 = 10 \text{ k}\Omega$, and $R_L = 1 \text{ k}\Omega$.

- (a) If the loop gain is large, what do you expect the closed-loop gain to be? Give both an expression and its value.
- (b) Find the dc emitter current in each of Q_1 and Q_2 . Also, find the dc voltage at the emitter of Q_2 .
- (c) Calculate the value of the loop gain $A\beta$. (Hint: Set $V_s = 0$ and break the loop at the base of Q_1 . Simplify the circuit by eliminating dc sources.)
- (d) Calculate the value of A_f .

D 11.30 Consider the series–shunt feedback amplifier of Fig. 11.8(c), which was the subject of Exercise 11.6. Assume that the voltage divider (R_1, R_2) is implemented with a 1-M Ω potentiometer. Assume that the MOSFET is biased so that $g_m = 4 \text{ mA/V}$ and r_o is large. Also, $R_D = 10 \text{ k}\Omega$. Find the value of R_1 that results in a closed-loop gain of 5 V/V.

D 11.31 Figure P11.31 shows a series–shunt feedback amplifier known as a “feedback triple.” All three MOSFETs are biased to operate at $g_m = 4 \text{ mA/V}$. You may neglect their r_o 's.

- (a) Select a value for R_F that results in a closed-loop gain that is ideally 10 V/V.

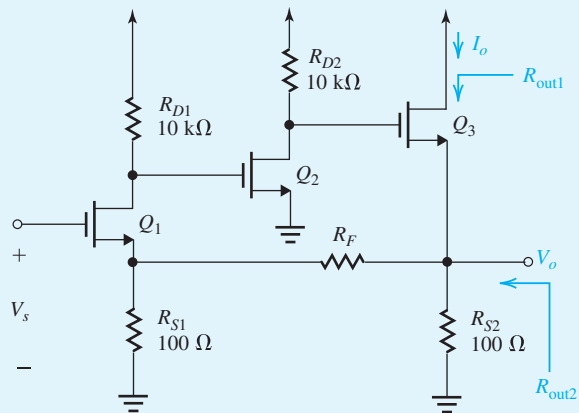


Figure P11.31

- (b) Determine the loop gain $A\beta$ and hence the value of A_f . By what percentage does A_f differ from the ideal value you designed for? How can you adjust the circuit to make A_f equal to 10?

11.32 Figure P11.32 shows a series–shunt feedback amplifier without details of the bias circuit.

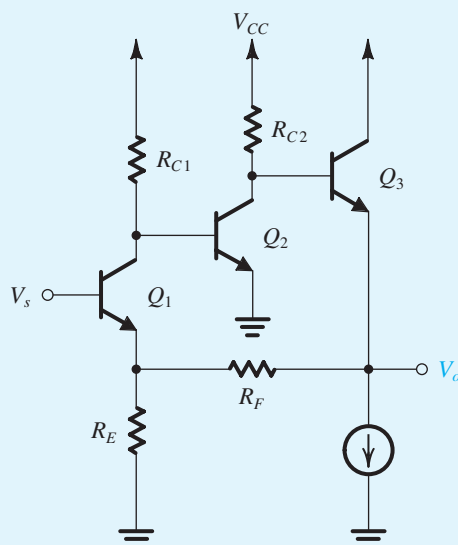


Figure P11.32

- (a) If R_E is selected to be $50\ \Omega$, find the value for R_F that results in a closed-loop gain with an ideal value of $25\ \text{V/V}$.
- (b) If Q_1 is biased at $1\ \text{mA}$, Q_2 at $2\ \text{mA}$, and Q_3 at $5\ \text{mA}$, and assuming that the transistors have $h_{fe} = 100$ and large r_o , and that $R_{C1} = 2\ \text{k}\Omega$ and $R_{C2} = 1\ \text{k}\Omega$, find the value of the loop gain $A\beta$ and hence of the closed-loop gain A_f .

D 11.33 The current-mirror-loaded differential amplifier in Fig. P11.33 has a feedback network consisting of the voltage divider (R_1, R_2), with $R_1 + R_2 = 1\ \text{M}\Omega$. The devices are sized to operate at $|V_{OV}| = 0.2\ \text{V}$. For all devices, $|V_A| = 10\ \text{V}$. The input signal source has a zero dc component.

- (a) Find the loop gain $A\beta$ and hence the value of A .
- (b) Find the values of R_1 and R_2 that result in a closed-loop gain of exactly $5\ \text{V/V}$.

Section 11.4: Systematic Analysis of Feedback Voltage Amplifiers (Series–Shunt)

11.34 A series–shunt feedback amplifier employs a basic amplifier with input and output resistances each of $2\ \text{k}\Omega$ and gain $A = 1000\ \text{V/V}$. The feedback factor $\beta = 0.1\ \text{V/V}$. Find

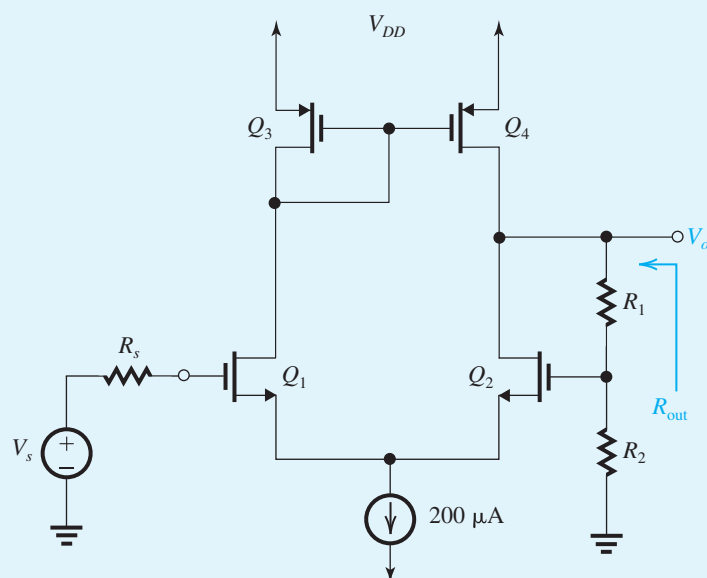


Figure P11.33

the gain A_f , the input resistance R_{if} , and the output resistance R_{of} of the closed-loop amplifier.

11.35 For a particular amplifier connected in a feedback loop in which the output voltage is sampled, measurement of the output resistance before and after the loop is connected shows a change by a factor of 200. Is the resistance with feedback higher or lower? What is the value of the loop gain $A\beta$? If R_{of} is 100 Ω , what is R_o without feedback?

11.36 The formulas for R_{if} and R_{of} in Eqs. (11.20) and (11.23), respectively, also apply for the case in which A is a function of frequency. In this case, the resulting impedances Z_{if} and Z_{of} will be functions of frequency. Consider the case of a series–shunt amplifier that has an input resistance R_i , an output resistance R_o , and open-loop gain $A = A_0/[1 + (s/\omega_H)]$, and a feedback factor β that is independent of frequency. Find Z_{if} and Z_{of} and give an equivalent circuit for each, together with the values of all the elements in the equivalent circuits.

11.37 A feedback amplifier utilizing voltage sampling and employing a basic voltage amplifier with a gain of 1000 V/V and an input resistance of 1000 Ω has a closed-loop input resistance of 10 k Ω . What is the closed-loop gain? If the basic amplifier is used to implement a unity-gain voltage buffer, what input resistance do you expect?

11.38 Consider the noninverting op-amp circuit of Example 11.4 for the case $R_1 = \infty$ and $R_2 = 0$.

- What is the value of β , and what is the ideal value of the closed-loop gain?
- Adapt the expressions found in Example 11.4 to obtain expressions for A and $A\beta$ for this case.
- For $\mu = 10^4$, $R_{id} = 100$ k Ω , $R_s = 10$ k Ω , $r_o = 1$ k Ω , and $R_L = 2$ k Ω , find A , $A\beta$, A_f , R_{in} , and R_{out} .

***11.39** This problem deals with the series–shunt feedback amplifier of Fig. P11.29 and overlaps somewhat with Problem 11.29. Thus, if you have already solved 11.29, you can use some of the results in the solution of this problem. The devices operate with $V_{BE} = 0.7$ V and have $\beta_1 = \beta_2 = 100$. The input signal V_s has a zero dc component. Resistances $R_s = 100$ Ω , $R_1 = 1$ k Ω , $R_2 = 10$ k Ω , and $R_L = 1$ k Ω .

- If the loop gain is large, what do you expect the closed-loop gain V_o/V_s to be? Give both an expression and its approximate value.

- Find the dc emitter current in each of Q_1 and Q_2 . Also find the dc voltage at the emitter of Q_2 .
- Sketch the A circuit without the dc sources. Derive expressions for A , R_i , and R_o , and find their values.
- Give an expression for β and find its value.
- Find the closed-loop gain V_o/V_s , the input resistance R_{in} , and the output resistance R_{out} . By what percentage does the value of A_f differ from the approximate value found in (a)?

SIM D *11.40 Figure P11.40 shows a series–shunt amplifier with a feedback factor $\beta = 1$. The amplifier is designed so that $v_o = 0$ for $v_s = 0$, with small deviations in v_o from 0 V dc being minimized by the negative-feedback action. The technology utilized has $k'_n = 2k'_p = 120$ $\mu\text{A/V}^2$, $|V_t| = 0.7$ V, and $|V'_A| = 24$ V/ μm .

- Show that the feedback is negative.
- With the feedback loop opened at the gate of Q_2 , and the gate terminals of Q_1 and Q_2 grounded, find the dc current and the overdrive voltage at which each of Q_1 to Q_5 is operating. Ignore the Early effect. Also find the dc voltage at the output.
- Find g_m and r_o of each of the five transistors.
- Find expressions and values of A and R_o . Assume that the bias current sources are ideal.
- Find the gain with feedback, A_f , and the output resistance R_{out} .
- How would you modify the circuit to realize a closed-loop voltage gain of 5 V/V? What is the value of output resistance obtained?

***11.41** Figure P11.41 shows a series–shunt amplifier in which the three MOSFETs are sized to operate at $|V_{ov}| = 0.2$ V. Let $|V_t| = 0.5$ V and $|V_A| = 10$ V. The current sources utilize single transistors and thus have output resistances equal to r_o .

- Show that the feedback is negative.
- Assuming the loop gain to be large, what do you expect the closed-loop voltage gain V_o/V_s to be approximately?
- If V_s has a zero dc component, find the dc voltages at nodes S_1 , G_2 , S_3 , and G_3 . Verify that each of the current sources has the minimum required dc voltage across it for proper operation.
- Find the A circuit. Calculate the gain of each of the three stages and the overall voltage gain, A .

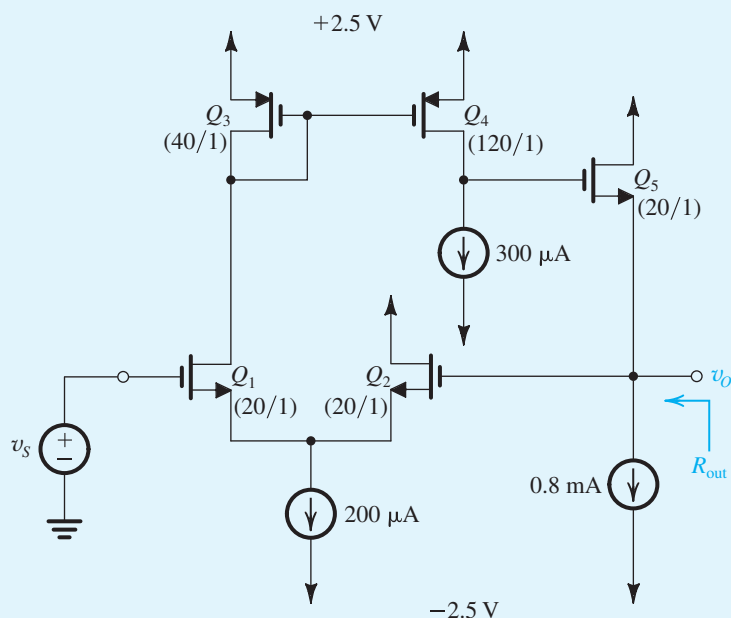


Figure P11.40

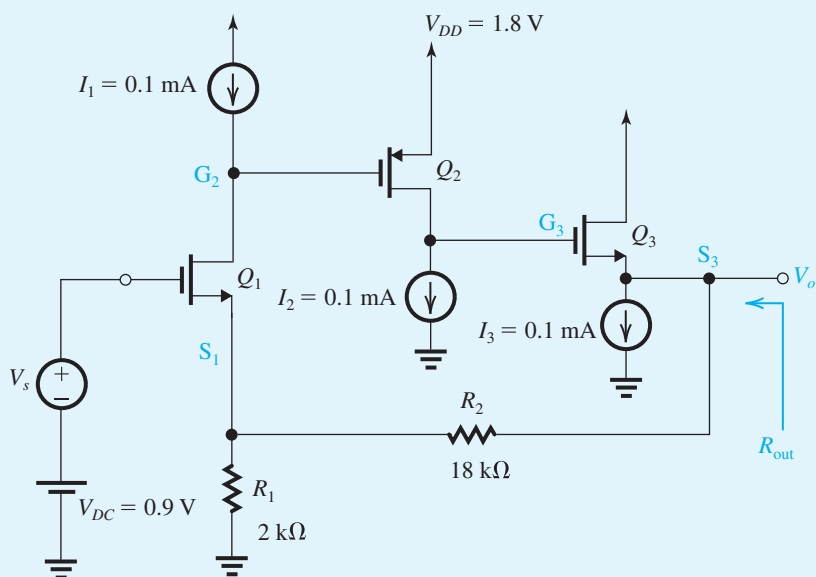


Figure P11.41

[Hint: A CS amplifier with a resistance R_s in the source lead has an effective transconductance $g_m/(1 + g_m R_s)$ and an output resistance $r_o(1 + g_m R_s)$.]

(e) Find β .

(f) Find $A_f = V_o/V_s$. By what percentage does this value differ from the approximate value obtained in (b)?

(g) Find the output resistance R_{out} .

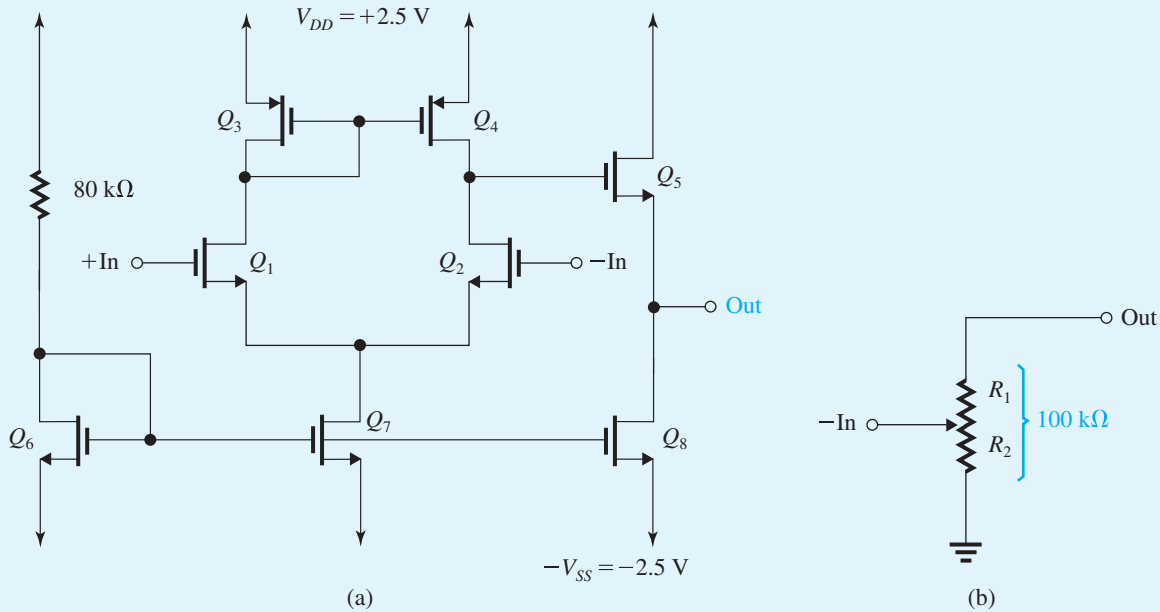


Figure P11.43

D *11.42 This problem deals with the series–shunt feedback amplifier of Fig. P11.33. Certain aspects of this amplifier were considered in Problem 11.33. If you have already solved problem 11.33, you will have the opportunity to compare results. The current-mirror-loaded differential amplifier has a feedback network consisting of the voltage divider (R_1, R_2), with $R_1 + R_2 = 1 \text{ M}\Omega$. The devices are sized to operate at $|V_{OV}| = 0.2 \text{ V}$. For all devices, $|V_A| = 10 \text{ V}$. The input signal source has a zero dc component.

- Show that the feedback is negative.
- What do you expect the dc voltage at the gate of Q_2 to be? At the output? (Neglect the Early effect.)
- Find the A circuit. Derive an expression for A and find its value.
- Select values for R_1 and R_2 to obtain a closed-loop voltage gain $V_o/V_s = 5 \text{ V/V}$.
- Find the value of R_{out} .
- Utilizing the open-circuit, closed-loop gain (5 V/V) and the value of R_{out} found in (e), find the value of gain obtained when a resistance $R_L = 10 \text{ k}\Omega$ is connected to the output.
- As an alternative approach to (f) above, redo the analysis of the A circuit including R_L . Then utilize the values of

R_1 and R_2 found in (d) to determine β and A_f . Compare the value of A_f to that found in (f).

D **11.43 The CMOS op amp in Fig. P11.43(a) is fabricated in a $1\text{-}\mu\text{m}$ technology for which $V_{tn} = -V_{tp} = 0.75 \text{ V}$, $\mu_n C_{ox} = 2\mu_p C_{ox} = 100 \mu\text{A/V}^2$, and $V'_A = 10 \text{ V}/\mu\text{m}$. All transistors in the circuit have $L = 1 \mu\text{m}$.

- It is required to perform a dc bias design of the circuit. For this purpose, let the two input terminals be at zero volts dc and neglect channel-length modulation (i.e., let $V_A = \infty$). Design to obtain $I_{D1} = I_{D2} = 50 \mu\text{A}$, $I_{D5} = 250 \mu\text{A}$, and $V_o = 0$, and operate all transistors except for the source follower Q_5 at $V_{OV} = 0.25 \text{ V}$. Assume that Q_1 and Q_2 are perfectly matched, and similarly for Q_3 and Q_4 . For each transistor, find I_D and W/L .
- What is the allowable range of input common-mode voltage?
- Find g_m for each of Q_1 , Q_2 , and Q_5 .
- For each transistor, calculate r_o .
- The $100\text{-k}\Omega$ potentiometer shown in Fig. 11.43(b) is connected between the output terminal (Out) and the inverting input (–In) to provide negative

D 11.64 The circuit in Fig. P11.64 utilizes a voltage amplifier with gain μ in a shunt–shunt feedback topology with the feedback network composed of resistor R_F . In order to be able to use the feedback equations, you should first convert the signal source to its Norton representation. You will then see that all the formulas derived in Example 11.9 apply here as well.

- If the loop gain is very large, what approximate closed-loop voltage gain V_o/V_s is realized? If $R_s = 2\text{ k}\Omega$, give the value of R_F that will result in $V_o/V_s \simeq -10\text{ V/V}$.
- If the amplifier μ has a dc gain of 10^3 V/V , an input resistance $R_{id} = 100\text{ k}\Omega$, and an output resistance $r_o = 2\text{ k}\Omega$, find the actual V_o/V_s realized. Also find R_{in} and R_{out} (indicated on the circuit diagram). You may use formulas derived in Example 11.9.
- If the amplifier μ has an upper 3-dB frequency of 1 kHz and a uniform -20-dB/decade gain rolloff, what is the 3-dB frequency of the gain $|V_o/V_s|$?

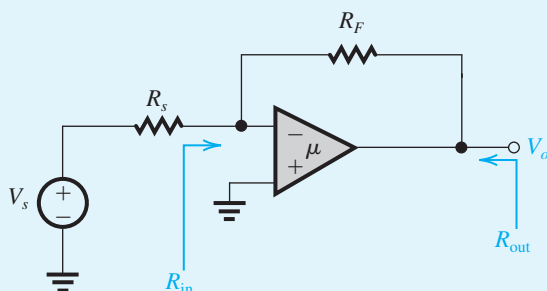


Figure P11.64

11.65 The feedback transresistance amplifier in Fig. P11.65 utilizes two identical MOSFETs biased by ideal current sources $I = 0.4\text{ mA}$. The MOSFETs are sized to operate at $V_{OV} = 0.2\text{ V}$ and have $V_t = 0.5\text{ V}$ and $V_A = 16\text{ V}$. The feedback resistance $R_F = 10\text{ k}\Omega$.

- If I_s has a zero dc component, find the dc voltage at the input, at the drain of Q_1 , and at the output.
- Find g_m and r_o of Q_1 and Q_2 .

- Provide the A circuit and derive an expression for A in terms of g_{m1} , r_{o1} , g_{m2} , r_{o2} , and R_F .
- What is β ? Give an expression for the loop gain $A\beta$ and the amount of feedback $(1 + A\beta)$.
- Derive an expression for A_f .
- Derive expressions for R_i , R_{in} , R_o , and R_{out} .
- Evaluate A , β , $A\beta$, A_f , R_i , R_o , R_{in} , and R_{out} for the component values given.

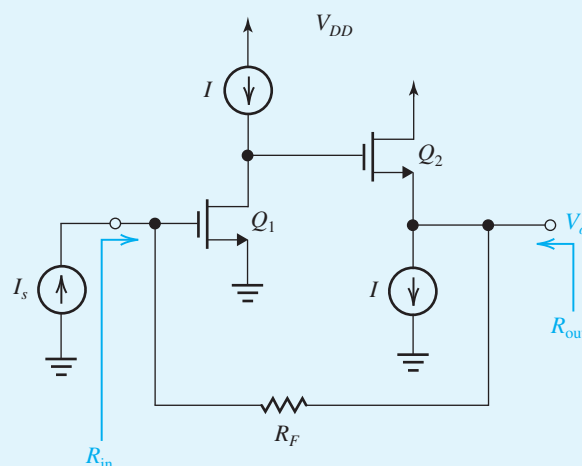


Figure P11.65

11.66 By setting $I_s = 0$ and breaking the feedback loop, find the loop gain of the feedback amplifier in Fig. P11.65. If you have already solved Problem 11.65, compare results. Which result do you think is more accurate, and why? For the numerical values given in Problem 11.65, by how much (in percent) do the two values of loop gain differ?

11.67 Analyze the circuit in Fig. E11.19 from first principles (i.e., do not use the feedback approach) and hence show that

$$A_f \equiv \frac{V_o}{I_s} = - \frac{(R_s \parallel R_F) \left(g_m - \frac{1}{R_F} \right) (r_o \parallel R_F)}{1 + (R_s \parallel R_F) \left(g_m - \frac{1}{R_F} \right) (r_o \parallel R_F) / R_F}$$

Comparing this expression to the one given in Exercise 11.19, part (b), you will note that the only difference is that g_m has been replaced by $(g_m - 1/R_F)$. Note that $-1/R_F$ represents the forward transmission in the feedback network, which the feedback-analysis method neglects. What is the condition then for the feedback-analysis method to be reasonably accurate for this circuit?

D 11.68 For the feedback amplifier in Fig. P11.68, select a value for R_F that results in a closed-loop gain $A_f \equiv V_o/I_s \simeq -10 \text{ k}\Omega$. Then, analyze the circuit to determine the actual value of A_f realized. As well, determine R_{in} and R_{out} . Transistors Q_1 and Q_2 are operated so that $g_{m1} = g_{m2} = 4 \text{ mA/V}$ and r_{o1} and r_{o2} can be neglected. Also, $R_{D1} = R_{D2} = 10 \text{ k}\Omega$.

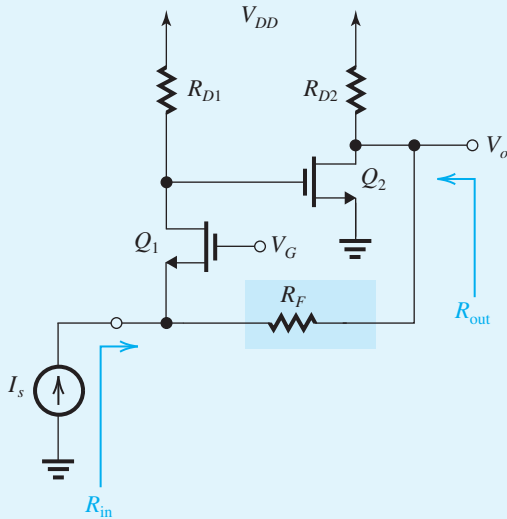


Figure P11.68

11.69 For the feedback transresistance amplifier in Fig. P11.69, let $V_{CC} = -V_{EE} = 5 \text{ V}$, $R_C = R_E = R_F = 10 \text{ k}\Omega$. The transistors have $V_{BE} = 0.7 \text{ V}$ and $\beta = 100$.

- (a) If I_s has a zero dc component, show that Q_1 and Q_2 are operating at dc collector currents of approximately 0.35 mA and 0.58 mA , respectively. What is the dc voltage at the output?

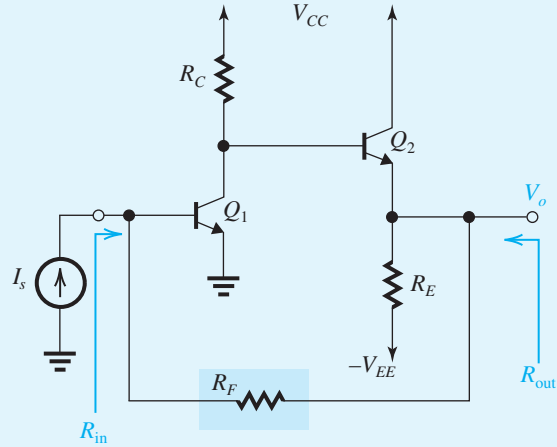


Figure P11.69

- (b) Find the A circuit and the value of A , R_i , and R_o . Neglect r_{o1} and r_{o2} .
 (c) Find the value of β , the loop gain, and the amount of feedback.
 (d) Find $A_f \equiv V_o/I_s$, the input resistance R_{in} , and the output resistance R_{out} .

D **11.70 (a) Show that for the circuit in Fig. P11.70(a), if the loop gain is large, the voltage gain V_o/V_s is given approximately by

$$\frac{V_o}{V_s} \simeq -\frac{R_f}{R_s}$$

- (b) Using three cascaded stages of the type shown in Fig. P11.70(b) to implement the amplifier μ , design a feedback amplifier with a voltage gain of approximately -100 V/V . The amplifier is to operate between a source resistance $R_s = 10 \text{ k}\Omega$ and a load resistance $R_L = 1 \text{ k}\Omega$. Calculate the actual value of V_o/V_s realized, the input resistance (excluding R_s), and the output resistance (excluding R_L). Assume that the BJTs have h_{fe} of 100. [Note: In practice, the three amplifier stages are not made identical, for stability reasons.]

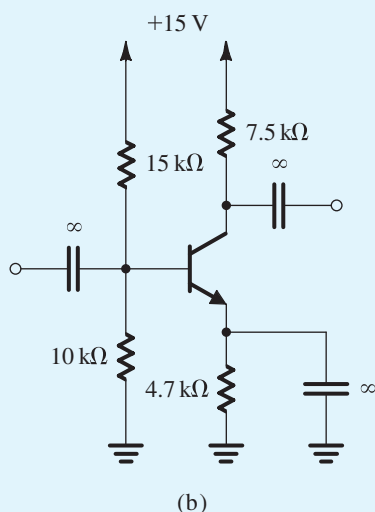
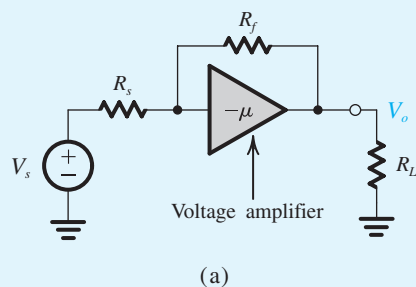


Figure P11.70

D 11.71 Negative feedback is to be used to modify the characteristics of a particular amplifier for various purposes. Identify the feedback topology to be used if:

- input resistance is to be lowered and output resistance raised.
- both input and output resistances are to be raised.
- both input and output resistances are to be lowered.

11.72 The feedback amplifier of Fig. P11.72 consists of a common-gate amplifier formed by Q_1 and R_D , and a feedback circuit formed by the capacitive divider (C_1 , C_2) and the common-source transistor Q_f . Note that the bias circuit for Q_f

is not shown. It is required to derive expressions for $A_f \equiv V_o/I_s$, R_{in} , and R_{out} . Assume that C_1 and C_2 are sufficiently small that their loading effect on the basic amplifier can be neglected. Also neglect r_o . Find the values of A_f , R_{in} , and R_{out} for the case in which $g_{m1} = 5 \text{ mA/V}$, $R_D = 10 \text{ k}\Omega$, $C_1 = 0.9 \text{ pF}$, $C_2 = 0.1 \text{ pF}$, and $g_{mf} = 2 \text{ mA/V}$.

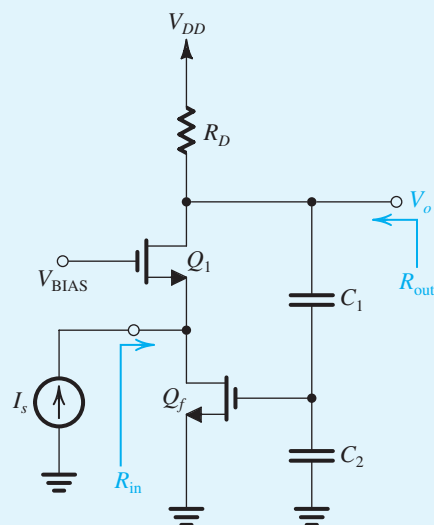


Figure P11.72

D *11.73 Figure P11.73 shows a shunt–shunt feedback amplifier. The MOSFETs have $V_m = 0.6 \text{ V}$, $V_A = 20 \text{ V}$, and $\mu_n C_{ox} = 200 \text{ }\mu\text{A/V}^2$. The power supply $V_{DD} = 3.3 \text{ V}$, and $R_L = 2 \text{ k}\Omega$. The coupling capacitor C_C can be assumed to be very large.

- Perform a dc design to meet the following specifications: $I_{D1} = 100 \text{ }\mu\text{A}$, $I_{D2} = 1 \text{ mA}$, $I_{R2,R1} = 10 \text{ }\mu\text{A}$, $V_{OV1} = V_{OV2} = 0.2 \text{ V}$. Neglect the Early effect. Specify the values required for I_1 , R_1 , R_2 , $(W/L)_1$, and $(W/L)_2$.
- Find an expression for β and hence an expression for the ideal value of V_o/V_s .
- Find the value of R_s that results in V_o/V_s being ideally -6 V/V .

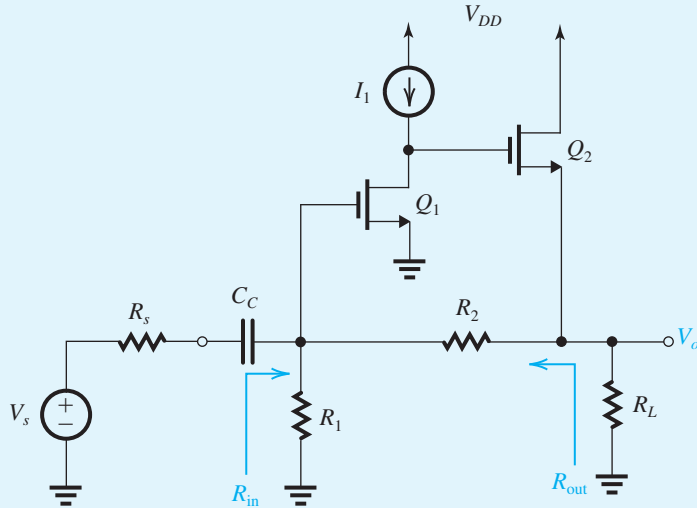


Figure P11.73

- (d) Find the A circuit and use it to determine the values of A , R_i , and R_o .
- (e) Find the value obtained for V_o/V_s .
- (f) Find R_{in} and R_{out} .

Feedback Current Amplifiers (Shunt–Series)

11.74 For the feedback current amplifier in Fig. P11.47:

- (a) Provide the A circuit and derive expressions for R_i and A . Neglect r_o of both transistors.
- (b) Provide the β circuit and an expression for β .
- (c) Find an expression for $A\beta$.
- (d) For $g_{m1} = g_{m2} = 5 \text{ mA/V}$, $R_D = 20 \text{ k}\Omega$, $R_M = 10 \text{ k}\Omega$, and $R_F = 90 \text{ k}\Omega$, find the values of A , β , $A\beta$, A_f , R_i , and R_{if} .
- (e) If $r_{o2} = 20 \text{ k}\Omega$ and $R_L = 1 \text{ k}\Omega$, find the output resistance as seen by R_L .

D 11.75 Design the feedback current amplifier of Fig. 11.27(a) to meet the following specifications:

- (i) $A_f \equiv I_o/I_s = -100 \text{ A/A}$
- (ii) amount of feedback $\simeq 40 \text{ dB}$
- (iii) $R_{in} \simeq 1 \text{ k}\Omega$

Specify the values of R_1 , R_2 , and μ . Assume that the amplifier μ has infinite input resistance and that $R_s = \infty$. For the

MOSFET, $g_m = 5 \text{ mA/V}$ and $r_o = 20 \text{ k}\Omega$. What R_{out} is obtained?

11.76 Consider the feedback current amplifier in Fig. 11.27(a) (which was analyzed in Example 11.10). Let $R_s = R_{id} = \infty$. By setting $I_s = 0$ and breaking the feedback loop at the gate of Q , find an expression for the loop gain $A\beta$. Evaluate $A\beta$ for the component values given in Example 11.10 and hence determine A and A_f . Why do the results differ somewhat from those found in Example 11.10?

11.77 The feedback current amplifier in Fig. P11.77 utilizes two identical NMOS transistors sized so that at $I_D = 0.2 \text{ mA}$ they operate at $V_{OV} = 0.2 \text{ V}$. Both devices have $V_t = 0.5 \text{ V}$ and $V_A = 10 \text{ V}$.

- (a) If I_s has zero dc component, show that both Q_1 and Q_2 are operating at $I_D = 0.2 \text{ mA}$. What is the dc voltage at the input?
- (b) Find g_m and r_o for each of Q_1 and Q_2 .
- (c) Find the A circuit and the value of R_i , A , and R_o .
- (d) Find the value of β .
- (e) Find $A\beta$ and A_f .
- (f) Find R_{in} and R_{out} .

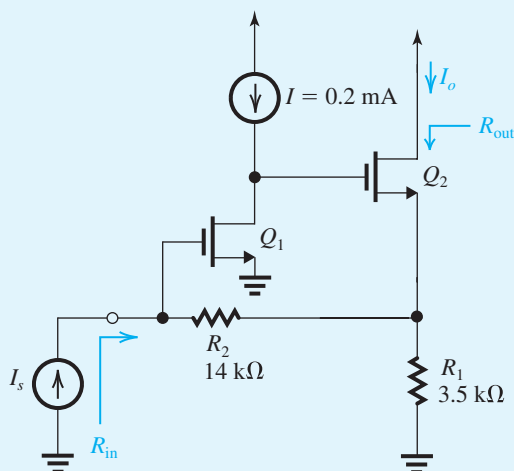


Figure P11.77

***11.78** The feedback current amplifier in Fig. P11.78(a) can be thought of as a “super” CG transistor. Note that rather than connecting the gate of Q_2 to signal ground, an amplifier is placed between source and gate.

- If μ is very large, what is the signal voltage at the input terminal? What is the input resistance? What is the current gain I_o/I_s ?
- For finite μ but assuming that the input resistance of the amplifier μ is very large, find the A circuit and derive expressions for A , R_i , and R_o .
- What is the value of β ?
- Find $A\beta$ and A_f . If μ is large, what is the value of A_f ?
- Find R_{in} and R_{out} assuming the loop gain is large.
- The “super” CG transistor can be utilized in the cascode configuration shown in Fig. P11.78(b), where V_G is a dc bias voltage. Replacing Q_1 by its small-signal model, use the analogy of the resulting circuit to that in Fig. P11.78(a) to find I_o and R_{out} .

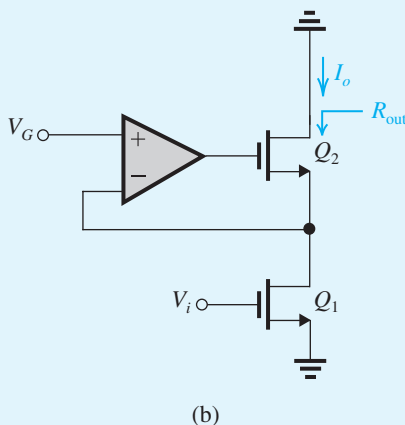
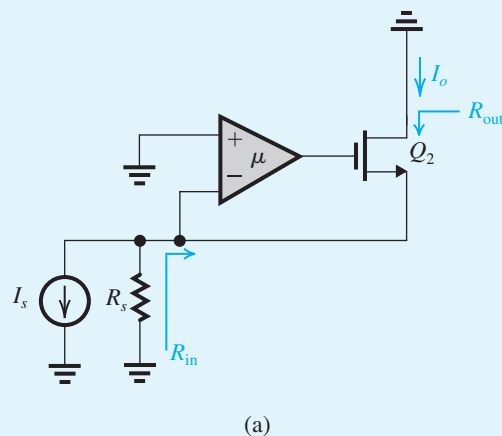


Figure P11.78

***11.79** Figure P11.79 shows an interesting and very useful application of feedback to improve the performance of the current mirror formed by Q_1 and Q_2 . Rather than connecting the drain of Q_1 to the gate, as is the case in simple current mirrors, an amplifier of gain $+\mu$ is connected between the drain and the gate. Note that the feedback loop does not include transistor Q_2 . The feedback loop ensures that the value of the gate-to-source voltage of Q_1 is such that I_{o1} equals I_s . This regulated V_{gs} is also applied to Q_2 . Thus, if W/L of Q_2

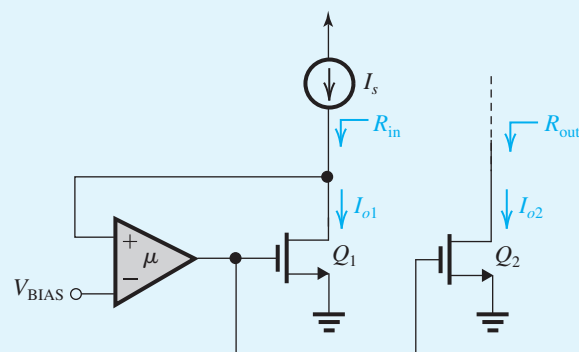


Figure P11.79

is n times W/L of Q_1 , $I_{o2} = nI_{o1} = nI_s$. This current tracking, however, is *not* regulated by the feedback loop.